IV. EDA-Data Inspection and Analysis

AIM:

- Viewing and inspecting DataFrames
- Filtering and subsetting data using conditions
- Descriptive statistics: measures of central tendency (mean, median, mode) and measures of dispersion (range, variance, standard deviation)

```python
import pandas as pd

# Reload data if needed
df = pd.read_csv('/content/test_Y3wMUE5_7gLdaTN.csv')

print("\n--- Columns ---")
print(df.columns)

categorical_cols = ['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed', 'Property_Area']
for col in categorical_cols:
    unique_vals = df[col].unique()
    print(f"\nUnique values in '{col}': {unique_vals}")

df['Dependents'] = df['Dependents'].replace('3+', 3)
df['Dependents'] = pd.to_numeric(df['Dependents'], errors='coerce').astype('Int64')

print("\nValue counts in 'Dependents' after conversion:")
print(df['Dependents'].value_counts(dropna=False))

cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']
for col in cols:
    print(f"\nStatistics for '{col}':")
    print(f"Mean: {df[col].mean():.2f}")
    print(f"Median: {df[col].median():.2f}")
    print(f"Mode: {df[col].mode().values}")
    print(f"Range: {df[col].max() - df[col].min():.2f}")
    print(f"Variance: {df[col].var():.2f}")
    print(f"Standard Deviation: {df[col].std():.2f}")
```

```
--- Columns ---
Index(['Loan_ID', 'Gender', 'Married', 'Dependents', 'Education',
       'Self_Employed', 'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
       'Loan_Amount_Term', 'Credit_History', 'Property_Area'],
      dtype='object')

Unique values in 'Gender': ['Male' nan 'Female']

Unique values in 'Married': ['No' 'Yes']

Unique values in 'Dependents': ['0' '3+' '2' '1' nan]

Unique values in 'Education': ['Not Graduate' 'Graduate']

Unique values in 'Self_Employed': ['Yes' 'No' nan]

Unique values in 'Property_Area': ['Urban' 'Semiurban' 'Rural']

Value counts in 'Dependents' after conversion:
Dependents
0       187
1        73
3        48
2        48
<NA>     11
Name: count, dtype: Int64

Statistics for 'ApplicantIncome':
Mean: 5429.25
Median: 5628.00
Mode: [1226 1685 2283 2449 4959 6110 6433 8544]
Range: 8908.00
Variance: 6993620.21
Standard Deviation: 2644.55
```

```
Statistics for 'CoapplicantIncome':
Mean: 436.38
Median: 0.00
Mode: [0]
Range: 959.00
Variance: 228684.40
Standard Deviation: 478.21

Statistics for 'LoanAmount':
Mean: 298.00
Median: 298.00
Mode: [298.]
Range: 0.00
Variance: 0.00
Standard Deviation: 0.00
```